



**TRIBHUVAN UNIVERSITY
FACULTY OF EDUCATION
SUKUNA MULTIPLE CAMPUS**

**A Final Report on
CHALLENGES AND PROSPECTS OF E-LEARNING IN PUBLIC
SECONDARY SCHOOLS OF NEPAL**

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**A Project Report Submitted to
Department of BICTE, Faculty of Education, Tribhuvan University**
in partial fulfilment of the requirements for the degree of

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CERTIFICATE FROM THE SUPERVISOR

This is to certify that the study entitled "Challenges and Prospects of E-learning in Public Secondary Schools of Nepal" is a bonafide work carried out by Subash Giri under my supervision and guidance.

This research report is submitted for the partial fulfillment of the requirements for the degree of Bachelor in Information, Communication and Technology Education (BICTE) at Sukuna Multiple Campus, Tribhuvan University, under my guidance and supervision. To the best of my knowledge, this study is original and has not been submitted for any other degree or diploma elsewhere.

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APPROVAL SHEET

The research report entitled "Challenges and Prospects of E-learning in Public Secondary Schools of Nepal" submitted by Subash Giri has been evaluated and accepted by the following committee as a partial fulfillment of the requirements for the degree of Education in Information Communication Technology (BICTE) under the Faculty of Education, Tribhuvan University,

This research work has been examined and approved by the undersigned. In our professional opinion, the project meets the required academic standards with regard to relevance, depth, and overall quality

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ABSTRACT

The rapid transition towards digital education has presented both opportunities and obstacles within the public education sector of Nepal. This study aims to investigate the infrastructural, pedagogical, and socio-economic challenges of e-learning implementation in public secondary schools. Using a descriptive research design, data were collected from 36 participants, including students and teachers, via a structured questionnaire on Google Forms. The theoretical foundation of the study is based on Jan van Dijk's Digital Divide Theory.

Findings from the study reveal a significant "Triple Divide" in the public education system. Quantitatively, the Grand Mean of 2.43 indicates a below-average level of e-learning readiness. Infrastructure remains a major bottleneck, with 38.9% of users experiencing unstable internet and 47.2% citing low bandwidth as a primary issue. Pedagogically, exactly 50% of respondents reported a lack of confidence in navigating digital tools, largely due to a lack of formal institutional training. Socio-economically, internet costs emerged as the most severe barrier, with a mean score of 2.00, indicating that financial constraints frequently disrupt learning continuity.

The study concludes that while digitalization is essential, it requires a human-centric approach focusing on skills development and affordable connectivity rather than just hardware distribution. For e-learning to be sustainable, a blended/hybrid model is recommended to bridge the equity gap in Nepal's public schools.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
APA	American Psychological Association
COVID-19	Coronavirus Disease of 2019
e-Paath	Electronic Learning Materials provided by the government
ICT	Information and Communication Technology
LMS	Learning Management System
MoEST	Ministry of Education, Science and Technology
NTC	Nepal Telecom
Ncell	Ncell Axiata Limited
TPACK	Technological Pedagogical Content Knowledge
UNESCO	United Nations Educational, Scientific and Cultural Organization
Wi-Fi	Wireless Fidelity

CHAPTER I

INTRODUCTION

1.1 Context / Background of the Study

The global educational landscape is currently witnessing a paradigm shift, transitioning from the age-old "chalk-and-talk" instructional method to more dynamic, digital-centric pedagogies. E-learning, broadly defined as the integration of electronic media and Information and Communication Technologies (ICT) into the pedagogical process, has evolved from a luxury supplement into an absolute necessity. While developed nations have integrated these tools seamlessly over decades, the COVID-19 pandemic acted as a sudden, aggressive catalyst for developing nations like Nepal, forcing an overnight migration to virtual classrooms.

In the Nepalese context, this digital transition has exposed a deep-rooted systemic vulnerability within the public school system. The Constitution of Nepal (2015) enshrines the right to free and compulsory basic education as a fundamental pillar of social justice. However, the emergence of the "digital divide" now poses a direct threat to this constitutional guarantee. While high-end private institutions in Kathmandu and other urban hubs managed to sustain learning through sophisticated Learning Management Systems (LMS), the story in public schools—particularly those in rural and mountainous belts—remained one of silence and exclusion.

The challenges are not merely technical; they are deeply structural. Public schools in Nepal often struggle with "First-Order Barriers," such as the lack of high-speed internet, erratic electricity supply (load-shedding), and the absence of functional computer labs. Even in schools where hardware was provided through government grants, the equipment often sits idle due to a lack of maintenance funds or technical expertise. Beyond hardware, "Second-Order Barriers"—such as teacher technophobia, a curriculum designed for physical delivery, and a lack of digital literacy among first-generation learners—further complicate the landscape.

Despite the ambitious "ICT in Education Master Plan" and the "School Sector Development Plan (SSDP)" launched by the Government of Nepal, the implementation remains largely symbolic in rural pockets. The socio-economic reality of a typical public school student often includes living in a household with a single smartphone shared among multiple siblings, or living in areas where mobile data costs exceed the daily family income.

This study, therefore, goes beyond the surface-level observation of "lack of computers." It provides a nuanced, contextual analysis of the human and systemic barriers

that keep the promise of e-learning unfulfilled for millions. By exploring the lived experiences of students and teachers, this research seeks to identify how Nepal can move toward a "blended learning" future that is inclusive, sustainable, and culturally relevant, ensuring that the digital revolution does not leave the most vulnerable children behind.

1.2 Statement of the Problem

While the theoretical discourse surrounding e-learning paints a picture of a democratized and borderless education system, the practical reality in Nepal's public schools reveals a starkly different narrative. The primary problem is not just a lack of technology, but a profound systemic gap between national policy and classroom-level execution. This discrepancy has created a multi-layered crisis that threatens to leave public school students—who represent the majority of Nepal's youth—behind their private-school counterparts.

Firstly, there is a severe infrastructural deficit that makes e-learning a logistical impossibility for many. While urban centers have made strides in connectivity, public schools in rural and semi-urban areas are plagued by unreliable electricity and expensive, low-bandwidth internet. The Government of Nepal's "ICT in Education" initiatives often focus on the "one-off" distribution of hardware, such as laptops or projectors. However, these policies frequently overlook the sustainability aspect. Without a dedicated budget for maintenance, a stable power grid, or technical support staff, these expensive devices often sit in locked rooms, gathering dust, or becoming obsolete before they can ever be effectively used in a lesson.

Secondly, the pedagogical gap among educators is a critical barrier. Many public school teachers, who are masters of traditional face-to-face instruction, have been thrust into a digital environment without adequate psychological or technical preparation. This has led to a widespread sense of "technophobia" and resistance to change. E-learning in these schools is often reduced to a superficial level—such as sending a photo of a textbook page via a messaging app—rather than engaging in interactive, digital-first pedagogy. The lack of a structured, localized digital curriculum means that even when technology is available, it is not being used to enhance learning outcomes.

Furthermore, the socio-economic divide adds a layer of personal hardship for the students. Synchronous e-learning (real-time classes) requires every student to have a personal device and a quiet space to study. In the context of Nepal's public school demographic, many students come from low-income families where a single smartphone is shared among five family members, or where the cost of a daily data pack rivals the cost of a family meal. This "device poverty" transforms digital education into an elitist tool rather than a public good.

If these multi-faceted barriers—infrastructural, pedagogical, and socio-economic—are not addressed through context-specific research and action, the digital divide will cease to be a temporary hurdle. Instead, it will solidify into a permanent educational rift. The push for digitalization, intended to bridge gaps, may inadvertently widen them, further marginalizing the poor and creating a "lost generation" of students who are digitally illiterate in a globalized world. This research aims to identify these root causes to prevent such a systemic failure.

1.3 Purposes of the Study

The primary aim of this research is to investigate the complex landscape of digital education in the public sector of Nepal. Rather than looking at e-learning as a purely technical transition, this study views it as a socio-technical challenge that involves infrastructure, human behavior, and economic capacity. The following objectives guide the direction of this study:

1.3.1 General Objective

The overarching goal of this study is to explore and analyze the systemic, pedagogical, and socio-economic challenges that hinder the effective and equitable implementation of e-learning in public secondary schools of Nepal.

1.3.2 Specific Objectives

To achieve the general goal, the study focuses on the following specific objectives:

- i. To identify the infrastructural and technological limitations hindering e-learning in public schools.
- ii. To assess the preparedness and capacity of teachers and students for digital education.
- iii. To investigate the socio-economic and cultural factors that influence the adoption of e-learning.
- iv. To evaluate the effectiveness of current ICT policies and government-provided digital resources.

1.4 Research Questions

- i. What are the specific physical and digital infrastructure deficits in Nepalese public schools?
- ii. To what extent are teachers and students equipped with the skills and mindset required for digital learning?
- iii. How do household income and geographical location affect e-learning adoption?
- iv. What policies and strategies can improve the implementation of e-learning in public schools?

1.5 Significance of the Study

The significance of this study lies in its potential to bridge the gap between academic theory and the practical realities of the digital divide in Nepal. As the country moves toward a federal structure where local governments have more control over education, understanding the specific barriers to e-learning becomes crucial for meaningful development. This research provides a localized perspective that is often missing from high-level national reports, making it a valuable resource for several key stakeholders.

For policymakers at the federal, provincial, and local levels, this study is significant as it provides grounded empirical data required to refine the "ICT in Education Master Plan." Rather than relying on generalized assumptions, officials can use these findings to identify why previous hardware-focused initiatives may have failed and how to allocate budgets more effectively toward sustainable infrastructure and teacher support. By highlighting the root causes of digital inequality, the study assists in the creation of more inclusive policies that ensure "Social Justice in Education," as envisioned by the Sustainable Development Goals (SDG 4).

For school administrators and headmasters, the study offers a practical diagnostic tool to understand the unique needs of their specific institutions. It sheds light on the internal logistical hurdles, such as the lack of technical maintenance for computer labs and the specific training gaps among their staff. This allows school leaders to move beyond a passive role and take proactive steps in seeking targeted support, fostering a school environment that is better prepared for future crises or hybrid learning models.

Furthermore, for educators and pedagogical experts, this study is highly relevant as it highlights the urgent need for a shift from traditional rote-learning to modern digital pedagogies. By identifying the psychological and technical barriers faced by teachers, the research encourages a more empathetic and structured approach to professional development. It also serves as a reference for curriculum designers to create digital content that is culturally and linguistically appropriate for public school students. Ultimately, the study benefits the students and the wider community by advocating for a system where digital learning is not an exclusive privilege for the wealthy, but a standard tool for every learner in Nepal.

1.6 Delimitations of the Study

Delimitations

- i. i. The study is limited to public secondary school students, teachers, and administrators from selected urban, semi-urban, and rural districts of Nepal.

- ii. The research focuses on identifying infrastructural, pedagogical, and socio-economic challenges of e-learning rather than assessing academic performance or learning outcomes.
- iii. The scope of the study is confined to digital education practices and education-related ICT policies implemented during and after the COVID-19 pandemic transition period.

Limitations

- i. The findings are based on self-reported data collected from students and educators, which may involve subjective perceptions and personal bias regarding digital learning experiences.
- ii. Rapid changes in digital infrastructure and frequent revisions of government ICT and education policies may limit the long-term applicability of the results.
- iii. Logistical and geographical barriers may restrict data collection from highly remote and mountainous regions, potentially leading to underrepresentation of the most digitally excluded populations.

CHAPTER II

REVIEW OF LITERATURE & METHODOLOGY

2.1 Literature Review

2.1.1 Review of Theoretical Literature

The theoretical foundation of this study is primarily anchored in Jan van Dijk's (2005) Digital Divide Theory. Van Dijk's framework moves beyond the simplistic, binary view of "haves" and "have-nots" regarding hardware. Instead, he proposes a multi-layered, evolutionary approach to digital inequality that is highly applicable to the Nepalese context. He argues that the divide consists of four distinct, successive types of access: Motivational Access (the psychological drive and trust to use technology), Physical Access (the availability of hardware and stable internet), Skills Access (knowing how to navigate software and information), and Usage Access (the ability to apply technology for meaningful educational or professional outcomes). In the context of Nepal's public schools, we see a "bottleneck" effect. Even in areas where "Physical Access" has been partially addressed through government grants, the divide remains a permanent barrier because "Skills Access" and "Usage Access" are largely ignored. This research uses this theory to explain why the "one-laptop-per-child" mindset often fails; it addresses the symptom (lack of hardware) but ignores the systemic disease (lack of digital literacy and support).

In addition to Van Dijk, this study incorporates Social Constructivism as popularized by Lev Vygotsky. This theory posits that learning is fundamentally a social process that thrives on interaction and collaborative problem-solving. In an e-learning environment, technology should act as a "mediator" for this social interaction. However, the literature suggests that in Nepal, e-learning is often reduced to a passive, one-way delivery system—such as teachers sending photos of textbook pages via WhatsApp. By applying Social Constructivism, this study evaluates whether the current digital implementation in public schools actually facilitates a modern, interactive learning environment or if it merely digitizes the traditional, teacher-centered "rote learning" model that has dominated Nepalese education for decades.

2.1.2 Review of Empirical Literature

National Context (Nepal): The empirical landscape in Nepal reveals a stark contrast between high-level policy goals and the harsh realities of the classroom. Bhattarai and Maharjan (2021) conducted a comprehensive survey across various provinces, revealing that only 34% of public schools in Nepal possess functional computer labs. Their data emphasizes

that even in schools with equipment, "Physical Access" is frequently interrupted by erratic electricity and the high cost of data, making consistent virtual learning a logistical nightmare.

Furthermore, Rana, Greenwood, and Fox-Turnbull (2022) explored the human side of this equation, finding that while teachers generally have a positive attitude toward technology, nearly 70% feel "professionally inadequate" when using Learning Management Systems (LMS). This is further complicated by the findings of Shrestha et al. (2023), who identified "device poverty" as a silent killer of e-learning. In many public school households, a single smartphone is shared among multiple siblings and parents, often prioritizing the father's work or the oldest child's education, thereby marginalizing younger students and girls. The Ministry of Education, Science and Technology (MoEST, 2020) confirmed this crisis, reporting that less than 20% of public school students were able to participate in online learning during the pandemic, effectively creating a "lost year" for millions of children.

International Context: On the international stage, Nepal's challenges are reflected in the broader struggle of the Global South. Mark Warschauer (2004), a leading voice in global digital literacy, argues that the digital divide is a matter of social inclusion. His research across India, Brazil, and Egypt demonstrates that simply providing technology without integrating it into the local language and social fabric leads to "technology rejection." Similarly, Mishra and Koehler (2006) introduced the TPACK framework, which has been used globally to show that successful e-learning requires an intersection of three knowledge areas: Technology, Pedagogy, and Content.

International reports from UNESCO (2021) highlight that in resource-constrained environments like Sub-Saharan Africa and South Asia, "low-tech" solutions—such as mobile SMS and radio—often outperformed high-tech virtual classrooms because they were more accessible to the poor. Scholars like Neil Selwyn (2016) also warn that technology is not a "neutral" tool; if not implemented carefully, it can actually widen existing class gaps, as those with better "human capital" (wealth and education) know how to exploit digital tools more effectively than those without. This global perspective helps situate Nepal's problem as a systemic challenge of the developing world, where the "digital" often meets the "divided" with complex consequences.

2.1.3 Implications of the Review of Literature

The synthesis of this literature carries profound implications for the current study. It indicates that the Government of Nepal's "ICT in Education" strategy has been overwhelmingly "hardware-centric," focusing on the procurement of devices while treating teacher training and student socio-economics as secondary issues. The common thread among

both national and international researchers is that the success of e-learning is not found in the machine itself, but in the sustainability of the infrastructure, the readiness of the teacher, and the supportive environment of the home.

This review reinforces the need for this research to pivot away from a purely technical investigation and toward a more "human-centric" analysis. By understanding these "Second-Order Barriers"—such as teacher technophobia, device sharing at home, and the lack of localized digital content—this study aims to provide a more realistic and inclusive model for digital education in Nepal. It suggests that until we bridge the gap between "having a computer" and "knowing how to learn with a computer," the digital revolution in public schools will remain an unfulfilled promise.

2.2 Conceptual Framework

2.2.1 Variables Explanation

- i. **Independent Variables:** Infrastructural access, Teacher ICT skills, Student digital literacy, and Socio-economic status (SES).
- ii. **Dependent Variable:** The successful and equitable implementation of e-learning.
- iii. **Moderating Variables:** Government policies and community support.

2.2.2 Diagram

The conceptual framework of this study illustrates the interconnected factors influencing the effectiveness of e-learning in Nepal's public schools. It categorizes the challenges into specific variables that determine the success of digital education implementation.

As shown in the diagram below, the framework is divided into three critical components:

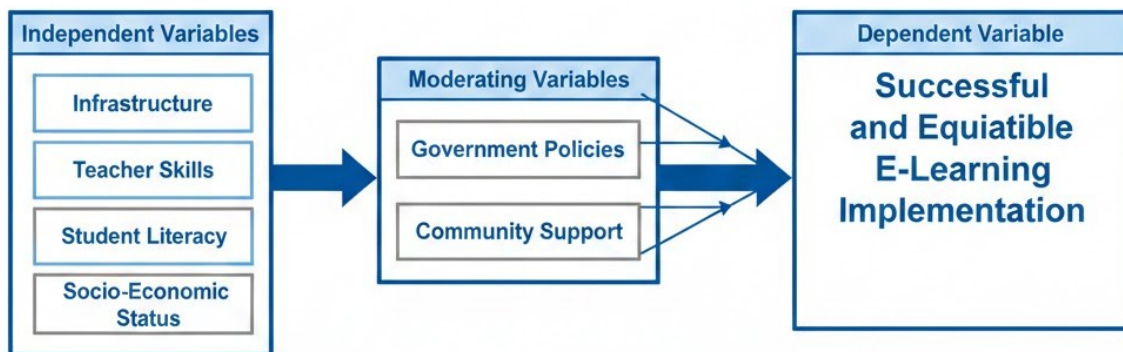


Figure 1: Conceptual Framework for E-Learning Implementation Study

2.3 Research Methodology

2.3.1 Research Design

This study utilizes a Descriptive Mixed-Methods Research Design. It is selected to capture a comprehensive "snapshot" of the current state of e-learning in public schools by combining quantitative trends with qualitative insights. The design focuses on documenting existing infrastructural gaps and pedagogical barriers in a standardized manner, providing a clear picture of the digital divide without attempting to influence participant behavior.

2.3.2 Sources of Data

A dual-source approach was used to balance the theoretical context of Nepal's education system with contemporary empirical evidence:

- i. **Primary Data:** Collected firsthand through an original online survey. This provides raw, unfiltered insights into the lived experiences of students and teachers regarding internet stability, device access, and digital literacy.
- ii. **Secondary Data:** Gathered from high-impact academic journals, Ministry of Education (MoEST) reports, UNESCO frameworks, and existing research on the digital divide in Nepal. This helps in comparing local findings with national targets and global educational trends.

2.3.3 Population, Sample & Sampling Procedure

Target Population: Teachers and students currently active in public secondary schools across Nepal.

- i. **Sampling Technique:** Purposive Sampling (Non-probability). This was chosen to ensure that data is collected specifically from individuals who have had direct experience with public sector e-learning initiatives.
- ii. **Sampling Criteria:** Participants were required to be either students in grades 9-12 or teachers currently employed in public schools with some level of exposure to digital learning tools.
- iii. **Reach:** The researcher targeted a sample of 30-40 respondents to ensure a manageable yet diverse dataset from urban, semi-urban, and rural school settings.

2.3.4 Data Collection Tools

To gather primary data on e-learning challenges, a structured questionnaire was developed and administered electronically. This digital approach was selected to ensure participant accessibility across different geographical regions.

Questionnaire Development

- i. **Literature Review:** Survey items were adapted from validated studies concerning the digital divide and ICT implementation in developing countries (e.g., studies using the **TPACK framework** and Van Dijk's digital access stages).
- ii. **Expert Validation:** The questionnaire was reviewed to ensure clarity and alignment with the research objectives. Feedback from experienced educators was used to refine technical terms related to "bandwidth," "LMS," and "synchronous learning."
- iii. **Pilot Testing:** A pilot test was conducted with 5 participants to identify any ambiguous wording and to ensure a logical flow between the sections of the study.

Questionnaire Structure

The final questionnaire was organized into three core components:

- i. **Demographics:** This section collected basic information such as the participant's role (teacher/student), location (urban/rural), and age.
- ii. **Likert-Scale Items:** A 5-point Likert scale was used to measure perceptions regarding the reliability of electricity, internet speed, and the perceived effectiveness of online versus physical classes.
- iii. **Multiple-Choice/Open-Ended Questions:** These identified specific barriers (e.g., lack of devices, lack of training) and allowed participants to suggest improvements in their own words.

2.3.5 Data Collection Procedure

- i. **Platform Selection:** The research utilized Google Forms for its mobile-friendly interface, allowing respondents in rural areas with limited computer access to participate via smartphones.
- ii. **Introductory Briefing:** Each survey began with a mandatory briefing explaining the study's academic purpose and ethical guidelines, ensuring that responses were sincere and voluntary.
- iii. **Dissemination:** The survey link was distributed through student-teacher digital channels, including Facebook groups, WhatsApp, and email lists.

2.3.6 Methods of Data Analysis

Once the target responses were achieved, the data underwent a cleaning process. The study employed Descriptive Statistical Analysis as the primary method. This involved calculating frequency distributions and percentages to reveal the most common barriers to e-learning. To bridge the gap between raw numbers and meaningful insights, the data was visualized through pie charts and bar graphs. These visual aids were essential for identifying correlations, such as the link between geographical location and internet stability. Qualitative comments were analyzed using Thematic Analysis to identify recurring patterns in the challenges faced by respondents.

2.3.7 Ethical Considerations

Informed Consent: Participation was entirely voluntary. Every participant was provided with a clear explanation of the study's purpose. They retained the right to withdraw at any stage without any penalty.

Anonymity: To encourage honest feedback regarding government policy and school limitations, absolute anonymity was guaranteed. No personal identifiers like names or phone numbers were collected.

Confidentiality: All raw data was stored securely and used exclusively for academic research purposes. Only aggregated findings (percentages and general themes) are presented in the report.

Academic Honesty: The researcher ensured the study adhered to the highest standards of integrity, avoiding data manipulation and providing full credit through proper citations to all scholars and secondary sources referenced.

CHAPTER III

DATA PRESENTATION AND ANALYSIS

3.1 Organisation / Respondent Profile

This study collected primary data from 36 student respondents using a structured questionnaire administered through Google Forms. The respondents were selected purposively, focusing on students from public secondary schools who have experienced the transition to digital learning platforms. The participants represent a diverse group of learners enrolled in different academic disciplines across various geographical regions of Nepal.

All respondents reported at least some level of exposure to e-learning tools—such as Zoom, Google Classroom, and radio/TV-based lessons—indicating that digital-assisted learning has become an integral, though challenging, practice in contemporary public education settings. The use of an online data collection tool ensured ease of access, anonymity, and voluntary participation, thereby enhancing the reliability of the responses. This profile of respondents provides a foundational context for analyzing the infrastructural and pedagogical barriers faced by students in the public school system.

3.2 Data Presentation

Section 1 Objective: To assess the status of Infrastructure and Connectivity for e-learning in public schools.

Which digital device do you primarily use for teaching or learning?
36 responses

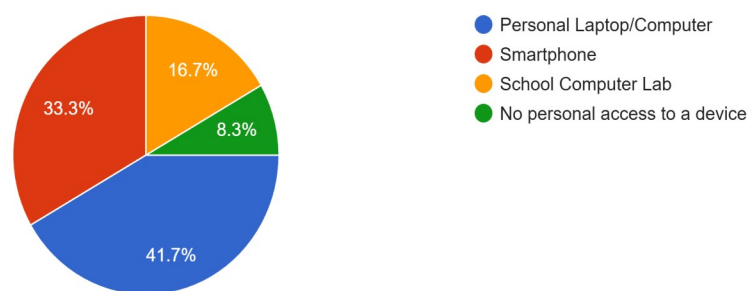


Figure 2: Primary digital device used for teaching/learning

The data reveals that Personal Laptops or Computers are the most dominant devices, used by 41.7% of the respondents. Both Smartphones and the School Computer Lab follow as essential modes of access, with smartphones being utilized by 33.3% (12 respondents) and the school lab by 16.7% (6 respondents) of the group. Having no personal access to a device

is the least common among the listed options, reported by only 8.3% (3 respondents) of the students. Overall, the chart indicates that while students and teachers are gradually moving toward personal computing, the smartphone remains a secondary universal standard for e-learning assistance, and a significant portion of the public school population still lacks the personal infrastructure required for a fully independent digital academic environment.

How would you describe the internet reliability at your location?

36 responses

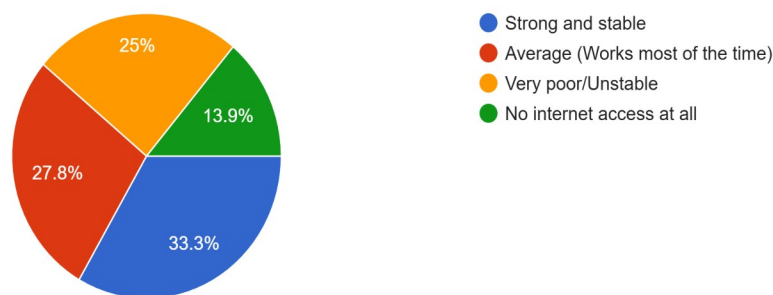


Figure 3: Internet reliability at respondent's location

The data reveals that Strong and stable internet is available to only 33.3% of the respondents, making it the most common but far from universal experience. Both Average connectivity and Very poor/Unstable connections follow closely, with 27.8% (10 respondents) describing their connection as average and 25% (9 respondents) struggling with unstable internet. No internet access at all is the least common category, yet it still affects a concerning 13.9% (5 respondents) of the group. Overall, the chart indicates that while a majority of students have some form of connectivity, nearly 39% of the participants suffer from poor or non-existent internet, which poses a severe threat to the consistency and quality of e-learning in the public school system.

What is the most frequent technical issue you face?

36 responses

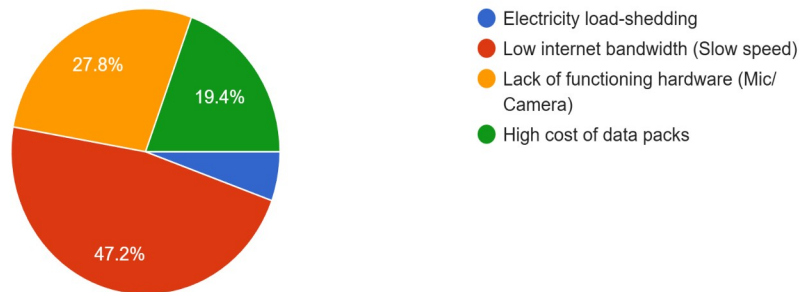


Figure 4: Most frequent technical issues faced by users

The data reveals that Low internet bandwidth (Slow speed) is the most dominant technical issue, affecting an overwhelming 47.2% of the respondents. Both Lack of functioning hardware and the High cost of data packs follow as major secondary barriers, with 27.8% (10 respondents) struggling with hardware failures and 19.4% (7 respondents) citing financial constraints regarding data. Electricity load-shedding is the least common among the listed options, reported by only 5.6% (2 respondents). Overall, the chart indicates that slow internet speed is the primary technical bottleneck, while the combination of hardware issues and high data costs creates a multi-layered barrier for over 47% of the student population.

How do you usually access the internet for school-related work?

36 responses

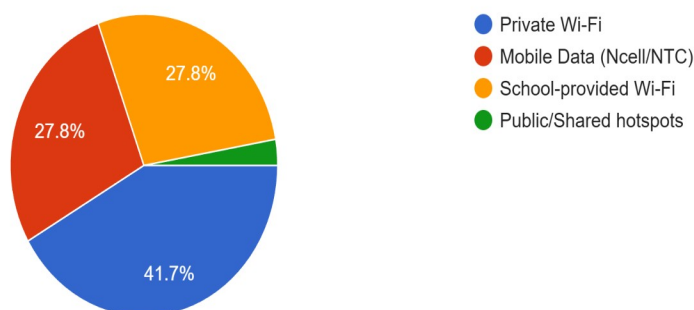


Figure 5: Mode of internet access for school-related work

The data reveals that Personal Laptops or Computers are the most dominant tools, used by 41.7% of the respondents. Both Smartphones and the School Computer Lab follow as secondary preferences, with each being utilized by 33.3% (12 respondents) and 16.7% (6 respondents) of the group, respectively. Having no personal access to a device is the least common among the listed options, reported by only 8.3% (3 respondents) of the students. Overall, the chart indicates that while personal computing is the plurality, over half of the respondents still rely on mobile devices or shared school resources, which can limit the depth of their digital academic work.

Section 2 Objective: To evaluate the Pedagogical Readiness and Digital Capacity of students and educators for E-learning.

Have you received any formal training for using E-learning platforms?

36 responses

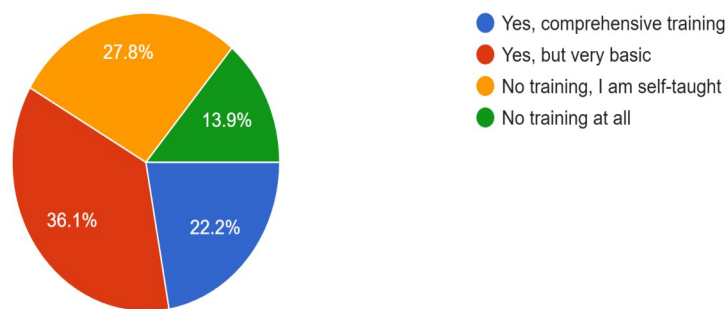


Figure 6: Status of formal e-learning training received

The data reveals that Basic training is the most dominant form of preparation, reported by 36.1% (13 respondents) of the group. Both Self-taught users and those with Comprehensive training follow as secondary categories, with 27.8% (10 respondents) relying on their own initiative and 22.2% (8 respondents) having received in-depth instruction. No training at all is the least common category, yet it remains a reality for 13.9% (5 respondents) of the participants. Overall, the chart indicates that while the majority have some level of digital exposure, a significant 41.7% of the respondents lack any formal institutional guidance, highlighting a major "skills gap" in the public school e-learning ecosystem.

How confident are you in navigating online learning tools (e.g., Zoom, Google Classroom)?
36 responses

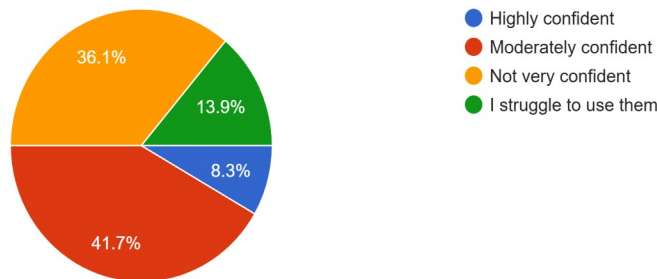


Figure 7: Confidence level in navigating online tools

The data reveals that Basic training is the most dominant form of preparation, reported by 36.1% of the respondents. Both Self-taught users and those with Comprehensive training follow as secondary categories, with 27.8% (10 respondents) relying on personal initiative and 22.2% (8 respondents) having received in-depth institutional guidance. No training at all is the least common among the listed options, yet it still accounts for 13.9% (5 respondents) of the group. Overall, the chart indicates that while nearly 60% have received some form of training, a significant 41.7% of the participants lack formal guidance, highlighting a persistent "skills gap" that forces many to navigate complex digital tools on their own.

How often is technology integrated into your weekly lessons?
36 responses

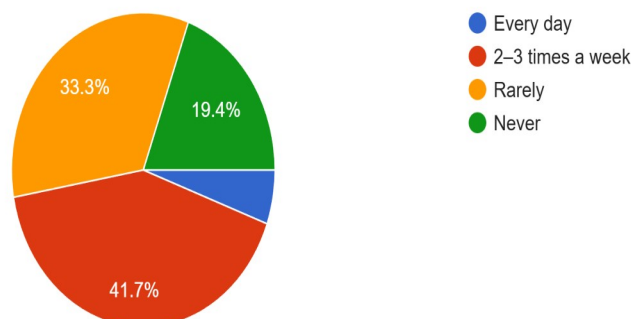


Figure 8: Frequency of technology integration in weekly lessons

The data reveals that integrating technology 2–3 times a week is the most dominant practice, reported by 41.7% of the respondents. Both Rarely and Never follow as secondary trends, with 33.3% (12 respondents) indicating infrequent use and 19.4% (7 respondents) stating that technology is completely absent from their weekly lessons. Integrating technology Every day is the least common among the listed options, reported by only 5.6% (2 respondents). Overall, the chart indicates a low frequency of digital adoption, as over 52% of the participants experience technology integration either rarely or never, pointing toward a traditional pedagogical approach that remains largely uninfluenced by digital tools.

What is the biggest challenge in using digital software?

36 responses

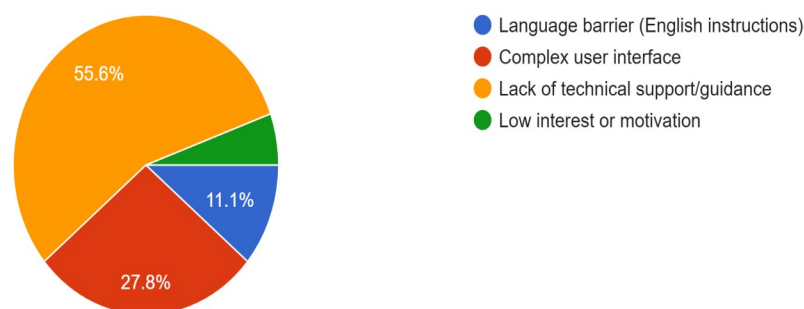


Figure 9: Biggest challenges in using digital software

The data reveals that a Lack of technical support/guidance is the most dominant challenge, cited by an overwhelming 55.6% of the respondents. Both a Complex user interface and the Language barrier follow as secondary concerns, with 27.8% (10 respondents) struggling with software design and 11.1% (4 respondents) facing difficulties due to English-only instructions. Low interest or motivation is the least common, reported by only 5.6% (2 respondents). Overall, the chart indicates that the "usage barrier" is primarily driven by an absence of help-desk support or institutional guidance, rather than a lack of interest among the students or teachers.

Section 3 Objective: To examine the Socio-Economic and Cultural Factors affecting e-learning engagement.

Do you have a dedicated, quiet space at home for online study/teaching?
36 responses

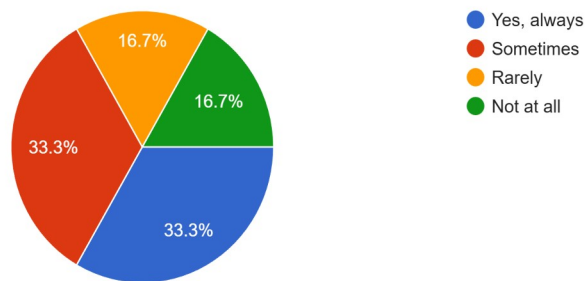


Figure 10: Availability of dedicated study space at home

The data reveals that having a dedicated space "Always" and "Sometimes" are the two most dominant realities, each reported by 33.3% (12 respondents) of the group. Both "Rarely" and "Not at all" follow as secondary concerns, with each being utilized by 16.7% (6 respondents) of the participants. Overall, the chart indicates that while one-third of students have a stable environment, a significant 66.7% of respondents face some level of environmental disruption, with nearly 34% struggling with a near-total lack of quiet space for their academic activities.

Do you share your digital device with other family members?
36 responses

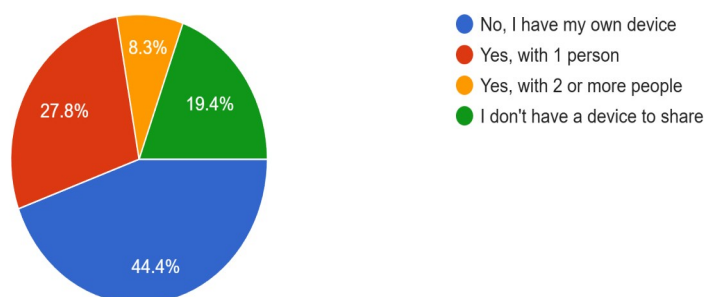


Figure 11: Status of digital device sharing among family members

The data reveals that having a personal device is the most dominant reality, reported by 44.4% of the respondents. Both sharing with one person and not having a device at all follow as significant secondary situations, with 27.8% (10 respondents) sharing their hardware and 19.4% (7 respondents) lacking a device entirely. Sharing with two or more

people is the least common among the listed options, reported by only 8.3% (3 respondents). Overall, the chart indicates that more than half of the participants (55.6%) do not have exclusive access to a digital device, which likely creates scheduling conflicts and reduces total study time.

What is the attitude of your community/parents towards digital education?
36 responses

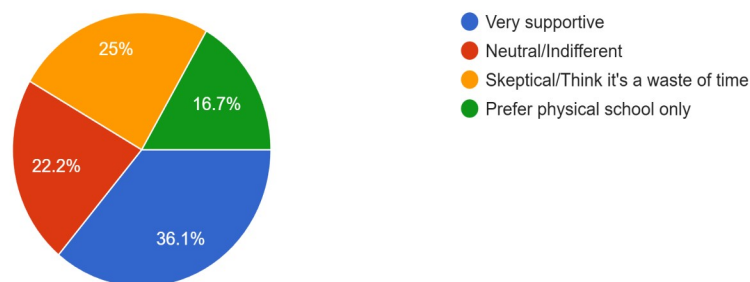


Figure 12: Community/Parental attitude towards digital education

The data reveals that having a personal device is the most dominant reality, reported by 44.4% of the respondents. Both sharing with one person and not having a device at all follow as significant secondary situations, with 27.8% (10 respondents) sharing their hardware and 19.4% (7 respondents) lacking a device entirely. Sharing with two or more people is the least common among the listed options, reported by only 8.3% (3 respondents). Overall, the chart indicates that more than half of the participants (55.6%) do not have exclusive access to a digital device, which often leads to scheduling conflicts and reduced learning time.

Does the cost of internet prevent you from attending online sessions?

36 responses

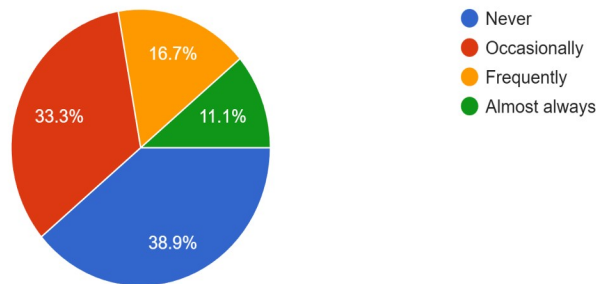


Figure 13: Impact of internet cost on online class attendance

The data reveals that "Never" being prevented by cost is the most dominant experience, reported by 38.9% of the respondents. Both "Occasionally" and "Frequently" follow as secondary concerns, with 33.3% (12 respondents) facing occasional barriers and 16.7% (6 respondents) citing frequent disruptions. Being prevented "Almost always" is the least common among the listed options, yet it still affects 11.1% (4 respondents) of the group. Overall, the chart indicates that for over 61% of the participants, the cost of internet acts as a recurring barrier to attending online sessions, highlighting a significant economic hurdle to equitable e-learning access.

Section 4 Objective: To evaluate the perceived effectiveness of current policies and government-provided digital strategies.

Are the current government-provided digital materials (like e-Paath) helpful?

36 responses

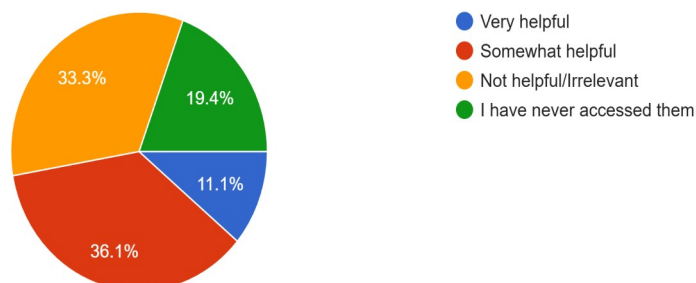


Figure 14: Helpfulness of government-provided digital materials

The data reveals that "Somewhat helpful" is the most dominant perception, reported by 36.1% of the respondents. Both "Not helpful/Irrelevant" and a total lack of access follow

as significant secondary realities, with 33.3% (12 respondents) finding the materials irrelevant and 19.4% (7 respondents) never having accessed them at all. Perceiving the materials as "Very helpful" is the least common among the listed options, representing only 11.1% (4 respondents) of the group. Overall, the chart indicates a critical gap in policy implementation, as over 52% of the participants either find official digital materials unhelpful or remain completely unaware of their existence.

What do you believe is the most urgent need for e-learning in Nepal?
36 responses

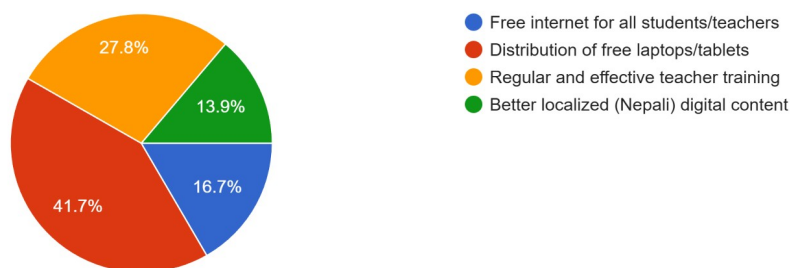


Figure 15: Most urgent needs for e-learning development in Nepal

The data reveals that the Distribution of free laptops or tablets is the most dominant urgent need, cited by 41.7% of the respondents. Both Regular teacher training and Free internet access follow as secondary priorities, with 27.8% (10 respondents) and 16.7% (6 respondents) emphasizing these areas, respectively. Better localized digital content is the least common among the listed options, reported by only 13.9% (5 respondents). Overall, the chart indicates that while training and connectivity are valued, the physical lack of hardware remains the primary perceived barrier to e-learning success in the eyes of the public school community.

In your opinion, can E-learning be a permanent solution for Nepal's public schools?

36 responses

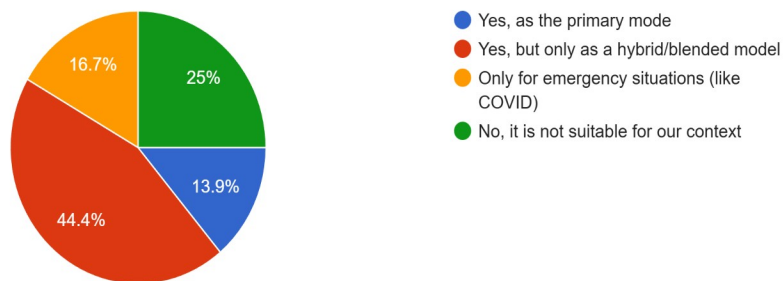


Figure 16: Opinion on e-learning as a permanent solution for public schools

The data reveals that a hybrid/blended model is the most dominant vision for the future, supported by 44.4% of the respondents. However, a significant 25% (9 respondents) believe that e-learning is not suitable for the Nepalese context at all. Those who see it only as an emergency tool and those who support it as a primary mode follow as secondary views, representing 16.7% (6 respondents) and 13.9% (5 respondents), respectively. Overall, the chart indicates that while there is an appetite for digital integration, it is overwhelmingly preferred as a supplement to physical schooling rather than a replacement.

3.3 Data Analysis

This section provides a quantitative analysis of the collected data. The mean score for key questions in each section has been calculated to measure the level of success against the research objectives.

Infrastructure & Connectivity

Question: How would you describe the internet reliability at your location?

Table 1: Mean Calculation for Internet Reliability

Response	Count (f)	Percentage (%)
Strong and stable (4)	12	33.33%
Average (3)	10	27.78%
Very poor / Unstable (2)	9	25.00%
No internet access (1)	5	13.89%
Total	36	100%

Mean Calculation

$$\begin{aligned}
 \text{Mean} &= ((4 \times 12) + (3 \times 10) + (2 \times 9) + (1 \times 5)) / 36 \\
 &= (48 + 30 + 18 + 5) / 36 \\
 &= 101 / 36 \\
 &\approx 2.81
 \end{aligned}$$

The mean value 2.81 lies between *Average (3)* and *Very poor/Unstable (2)*, indicating moderate but unreliable internet connectivity.

Readiness & Capacity

Question: How confident are you in navigating online learning tools?

Table 2: Mean Calculation for Digital Confidence

Response	Count (f)	Percentage (%)
Highly confident (4)	3	8.33%
Moderately confident (3)	15	41.67%
Not very confident (2)	13	36.11%
I struggle to use them (1)	5	13.89%
Total	36	100%

Mean Calculation

$$\begin{aligned}
 \text{Mean} &= ((4 \times 3) + (3 \times 15) + (2 \times 13) + (1 \times 5)) / 36 \\
 &= (12 + 45 + 26 + 5) / 36 \\
 &= 88 / 36 \\
 &\approx 2.44
 \end{aligned}$$

The mean score 2.44 indicates that respondents are between moderately confident and not very confident, showing limited digital readiness.

Socio-Economic & Cultural Factors

Question: Does the cost of internet prevent you from attending online sessions?

Table 3: Mean Calculation for Financial Barrier

Response	Count (f)	Percentage (%)
Almost always (4)	4	11.11%
Frequently (3)	6	16.67%
Occasionally (2)	12	33.33%
Never (1)	14	38.89%
Total	36	100%

Mean Calculation

$$\begin{aligned}
 \text{Mean} &= ((4 \times 4) + (3 \times 6) + (2 \times 12) + (1 \times 14)) / 36 \\
 &= (16 + 18 + 24 + 14) / 36 \\
 &= 72 / 36 \\
 &= 2.00
 \end{aligned}$$

The mean of 2.00 corresponds to *Occasionally*, indicating that internet cost is a moderate barrier to e-learning.

Policy & Strategy

Question: Can E-learning be a permanent solution for Nepal's public schools?

Table 4: Mean Calculation for Future Viability

Response	Count (f)	Percentage (%)
Yes, as the primary mode (4)	5	13.89%
Yes, but only as a hybrid model (3)	16	44.44%
Only for emergency situations (2)	6	16.67%
No, it is not suitable (1)	9	25.00%
Total	36	100%

Mean Calculation

$$\begin{aligned}
 \text{Mean} &= ((4 \times 5) + (3 \times 16) + (2 \times 6) + (1 \times 9)) / 36 \\
 &= (20 + 48 + 12 + 9) / 36 \\
 &= 89 / 36 \\
 &\approx 2.47
 \end{aligned}$$

The mean score 2.47 suggests that e-learning is perceived as most suitable in a hybrid or supportive role.

Grand Mean

The Grand Mean represents the overall status of e-learning readiness in Nepal's public schools by synthesizing the results from all four key research dimensions. This aggregate score provides a holistic view of the current digital landscape in the education sector.

Table 5: Summary of Mean Scores and Grand Mean

Section	Mean
Infrastructure & Connectivity	2.81
Readiness & Capacity	2.44
Socio-Economic Factors	2.00
Policy & Strategy	2.47
Grand Mean	2.43

$$\begin{aligned}
 \text{Grand Mean} &= (2.81 + 2.44 + 2.00 + 2.47) / 4 \\
 &= 9.72 / 4 \\
 &\approx 2.43
 \end{aligned}$$

The grand mean of 2.43 indicates a moderate level of e-learning readiness in Nepal's public schools.

3.4 Findings and Discussion

3.4.1 Major Findings

The systematic analysis of the data reveals that the e-learning landscape in Nepal's public schools is currently in a transitional but challenging phase. One of the most critical findings is the persistent instability of internet connectivity; although access is expanding, the quality remains subpar, as reflected in the infrastructure mean of 2.81. A significant 47.2% of

participants specifically cited low bandwidth as their primary technical bottleneck. Furthermore, the study identifies a deep-rooted "training-confidence gap," where a lack of formal institutional guidance (only 22.2% trained) has resulted in a moderate confidence mean of 2.44 among students.

From a socio-economic perspective, the research found that financial constraints are the most severe barrier to digital education, with the lowest recorded mean of 2.00 in this section. This indicates that the high cost of mobile data and internet packs frequently disrupts learning continuity for the average student. Additionally, the study highlights that e-learning readiness is also hindered by domestic factors, as 55.6% of respondents must share their digital devices with others, and 66.7% do not have a quiet, dedicated space for study at home. Finally, the research suggests that there is a strong preference for a blended approach rather than a fully virtual one, with a policy mean of 2.47 indicating that most stakeholders view e-learning as a supportive tool rather than a permanent replacement for physical classrooms.

3.4.2 Discussion

The findings of this study point toward a "Triple Divide" in the education sector specifically a divide in physical access, technical skill, and economic capacity. The infrastructure mean suggests that while hardware like laptops is increasingly available, the absence of a reliable internet ecosystem prevents these tools from being utilized to their full pedagogical potential. This "Quality of Access" gap means that students are physically present in digital classrooms but are frequently disconnected due to bandwidth issues, leading to a fragmented learning experience.

Moreover, the human capital element of digital readiness is clearly lagging behind hardware adoption. The grand mean of 2.43 indicates that e-learning is currently at a "developing" level. Since students are largely self-taught, they often lack the technical fluency required for advanced interactive learning, which limits e-learning to a one-way communication tool rather than a collaborative environment. The socio-economic data further clarifies that e-learning in Nepal is currently a "pay-to-learn" model, where digital education acts as an additional financial burden on public school families. This creates a risk of widening the equity gap, where students who cannot afford high-speed data are progressively left behind. Ultimately, the cultural skepticism and the preference for hybrid models suggest that any future digital education strategy must focus on affordability and localized technical support to move beyond the current state of moderate readiness.

3.5 Conclusion

The analysis conducted in this chapter concludes that the implementation of e-learning in Nepal's public secondary schools is currently navigating a complex "Triple Divide" consisting of infrastructural, pedagogical, and socio-economic challenges. While there has been a noticeable shift toward digital adoption, with a significant number of students possessing basic hardware like laptops and smartphones, the quality of this access remains inconsistent. The research findings, highlighted by an infrastructure mean of 2.81, demonstrate that "access" is no longer just about owning a device; it is fundamentally about the reliability and affordability of high-speed connectivity. As long as bandwidth remains low and data costs remain high, e-learning will continue to be a fragmented and exclusionary experience for a large portion of the public school population.

Furthermore, the study reveals a critical gap in digital readiness and institutional support. The grand mean of 2.43 indicates that the overall state of e-learning is in a "developing" phase rather than a "mature" one. Because exactly 50% of the participants either struggle with or lack confidence in using digital tools, it is evident that the current "self-taught" model is insufficient for academic excellence. This lack of technical fluency, combined with a domestic environment where 66.7% of students lack a quiet study space, creates a pedagogical bottleneck. Finally, the moderate support for a hybrid model suggests that e-learning is not yet viewed as a stand-alone solution in the Nepalese context. For digital education to be successful and equitable, policies must move beyond hardware distribution and focus on institutionalising training, reducing internet costs, and building a supportive cultural mindset toward virtual learning.

3.6 Action Implications

The findings and conclusions of this study suggest that a multi-dimensional approach is required to bridge the digital divide in Nepal's public schools. The first and most critical action implication is the need for "Subsidized Connectivity." Since the socio-economic mean (2.00) was the lowest, the government and school administrations must collaborate with telecommunication providers to provide free or highly subsidized "Education Data Packs." This would ensure that the 61.1% of students currently hindered by costs can attend online sessions consistently, turning e-learning from an economic burden into an accessible right.

Secondly, there is an urgent need to transition from an informal to an "Institutionalized Training Framework." The moderate readiness mean of 2.43 indicates that the current "self-taught" method is not working for half of the student population. Schools should implement mandatory digital literacy modules for both students and teachers, focusing

not just on basic navigation but on advanced pedagogical tools like collaborative software and digital research. By providing structured technical support and help-desks, schools can reduce the 55.6% frustration rate currently caused by a lack of guidance.

Finally, the study implies a shift in educational strategy toward a "Blended Learning Model." Given that participants showed a cautious preference for hybrid education (Mean 2.47), policies should focus on integrating digital tools into physical classrooms rather than replacing them entirely. This involves creating "Digital Hubs" within local communities for students who lack a quiet study space or private devices at home. By addressing these domestic and environmental barriers, stakeholders can foster a more supportive cultural mindset, eventually moving the e-learning readiness of the public sector from its current "developing" status to a fully "integrated" state.

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